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A Structured L1 Penalization Framework for Stable Sparse Power Modeling in Water Reuse Systems

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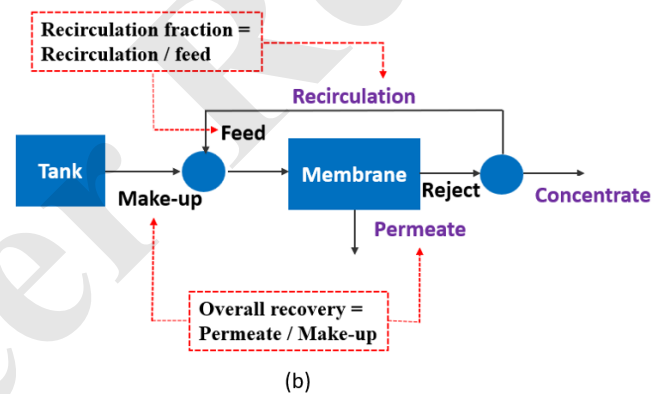
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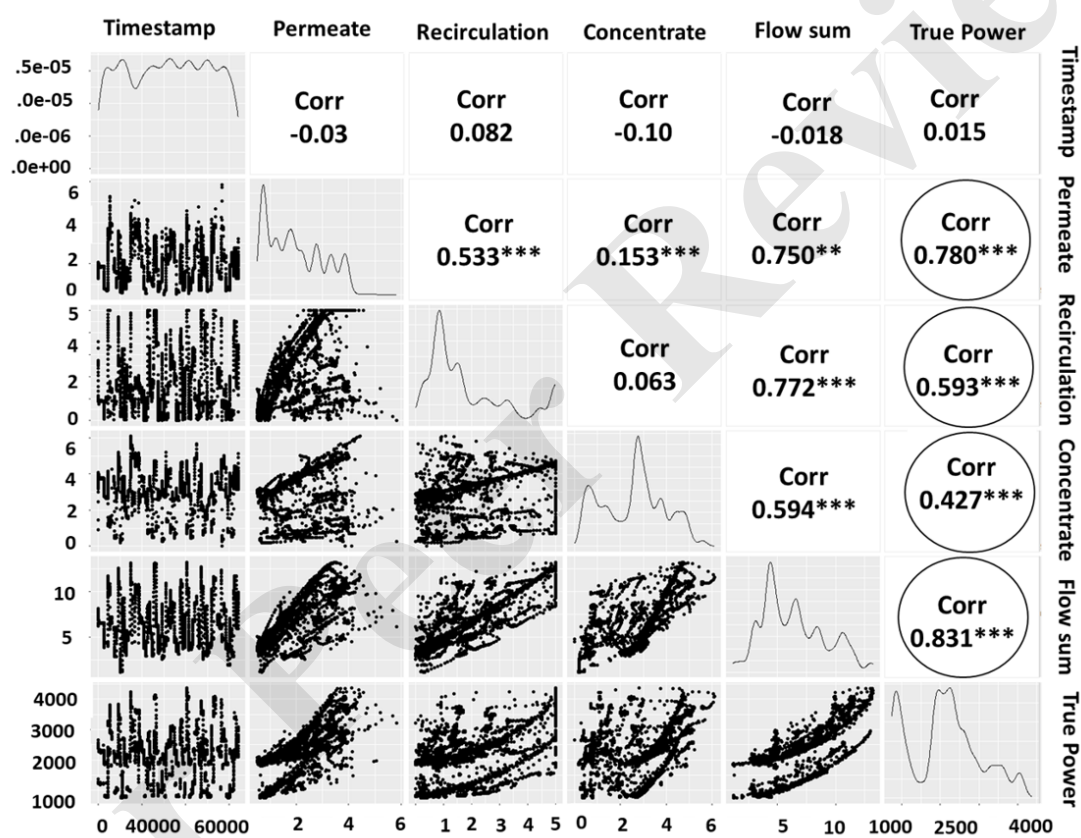
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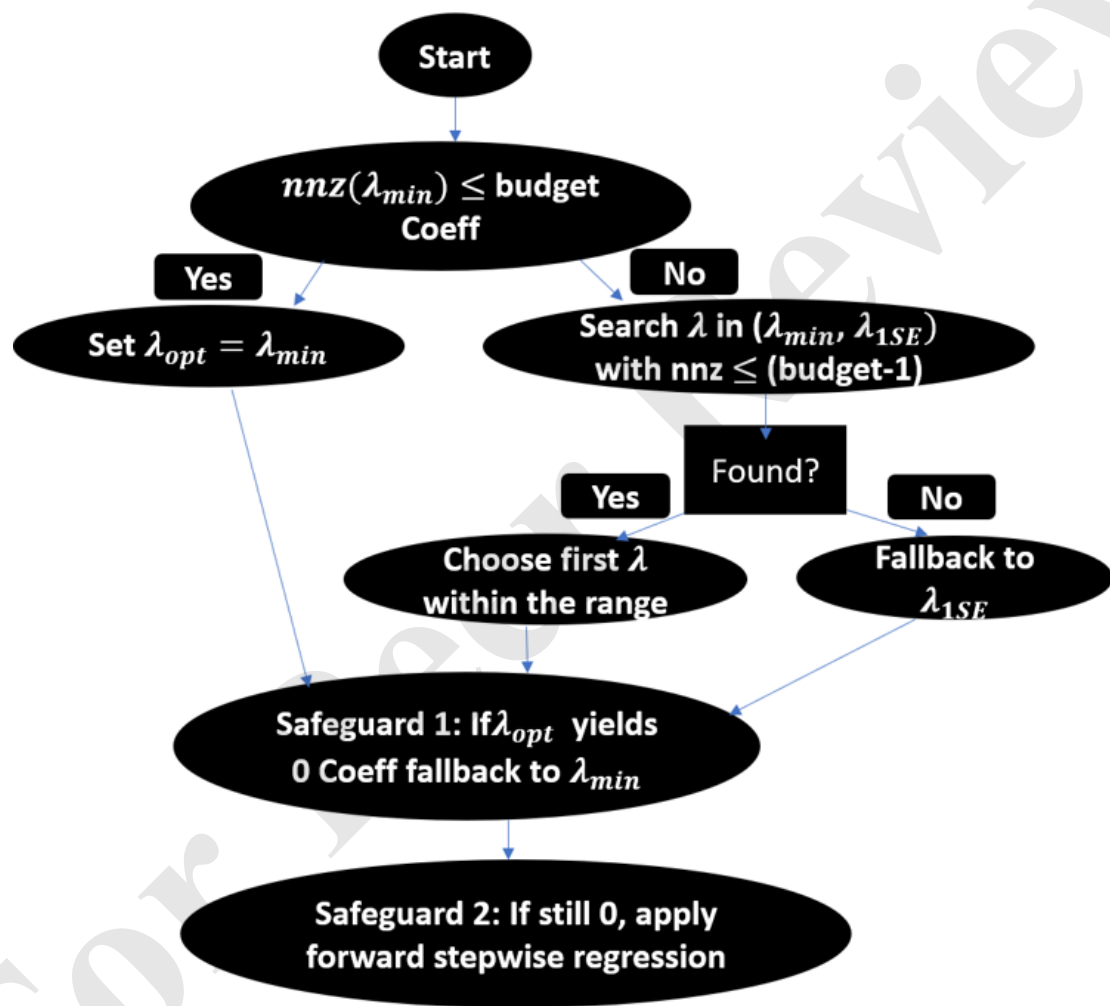


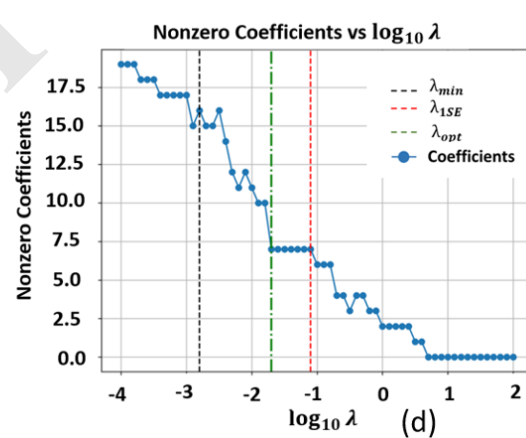
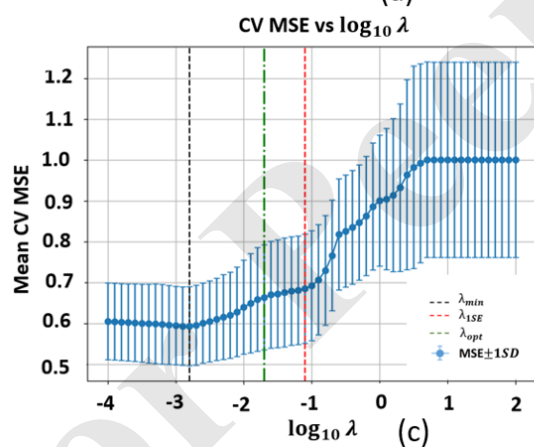
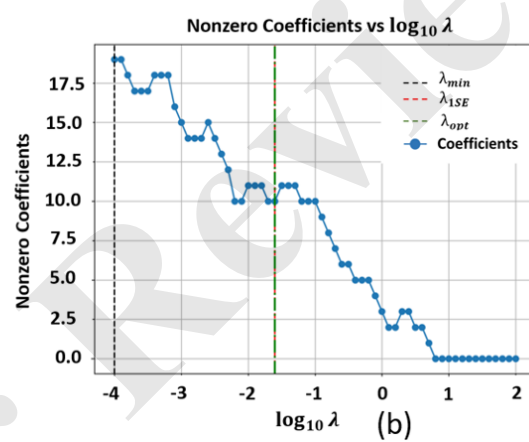
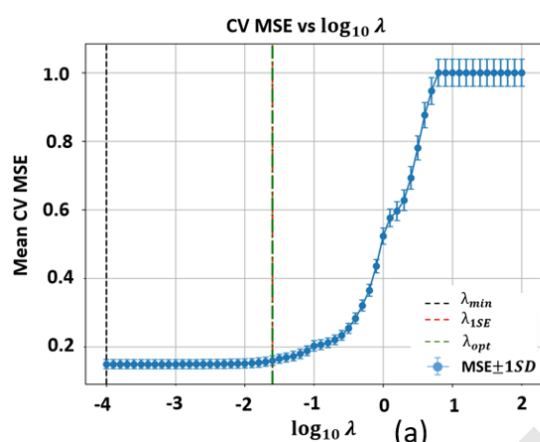
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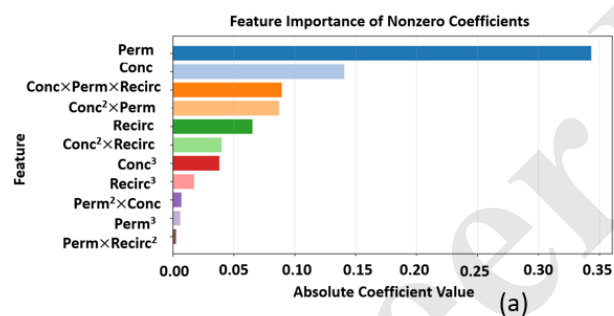


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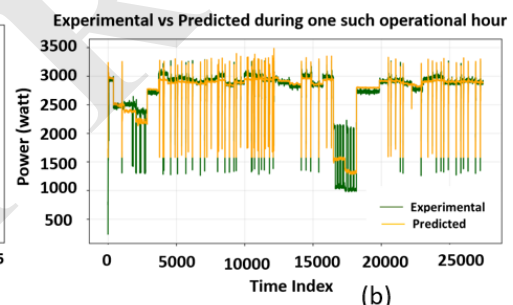




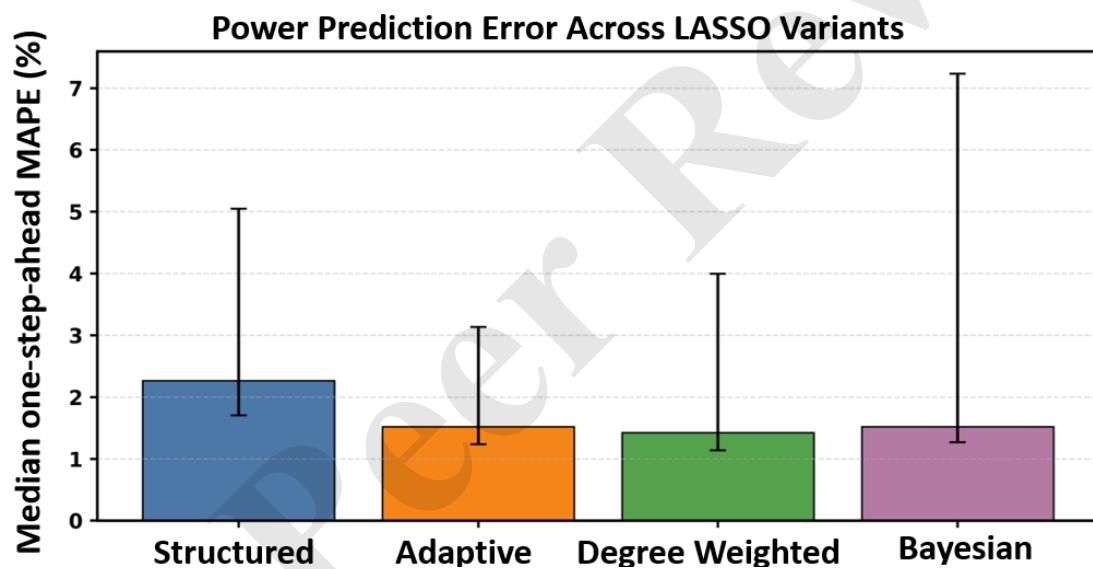


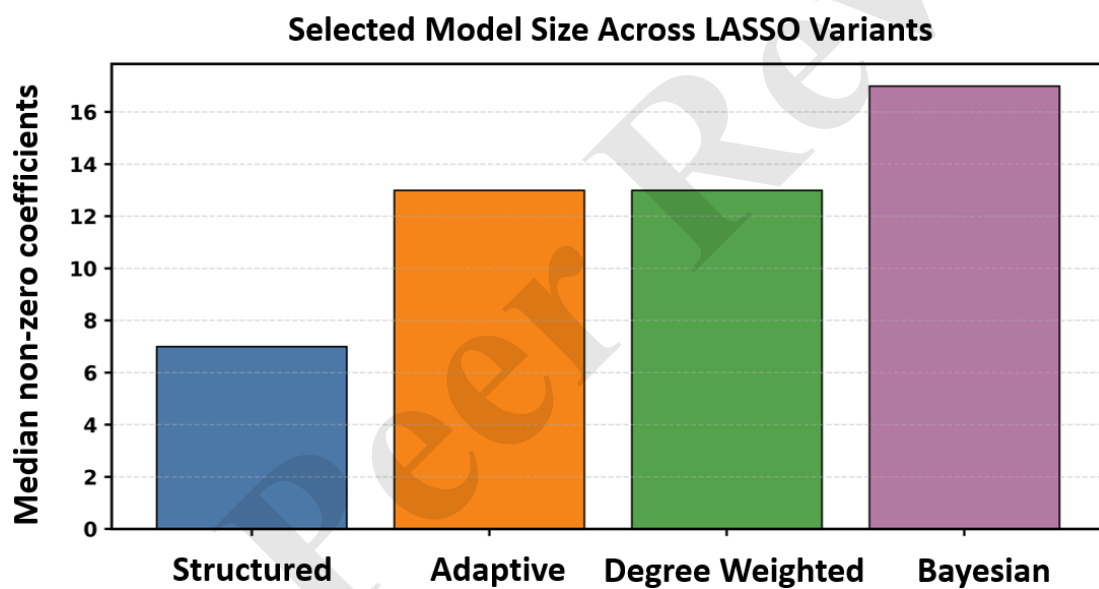


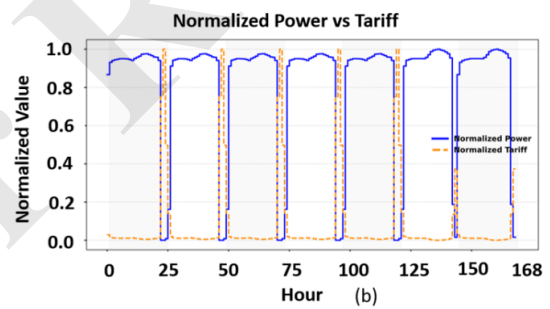
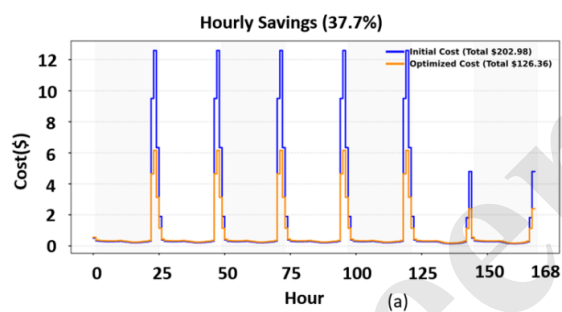
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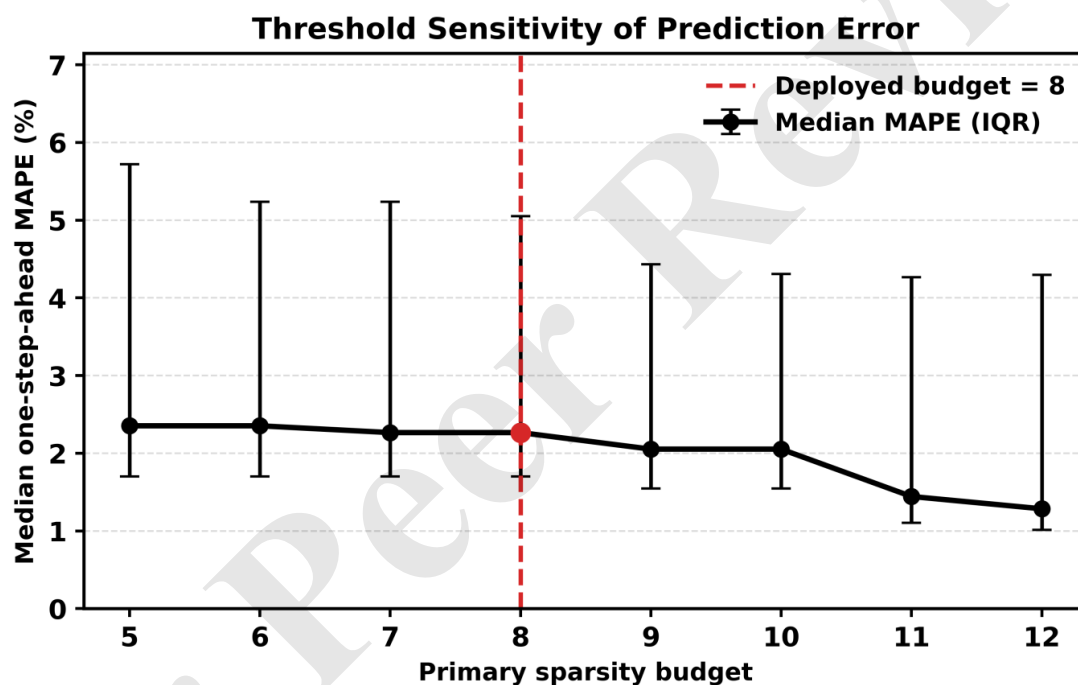


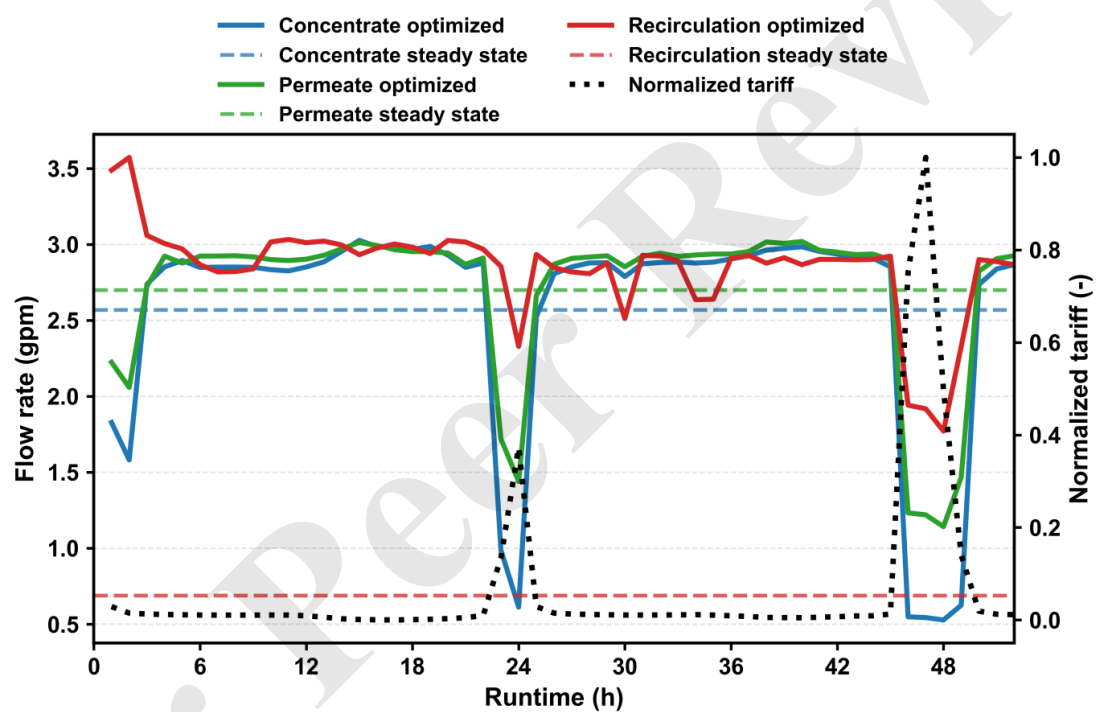
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A Structured L_1 Penalization Framework for Stable Sparse Power Modeling in Water Reuse Systems

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A Structured L_1 Penalization Framework for Stable Sparse Power Modeling in Water Reuse Systems

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Abstract. Water reuse and desalination facilities are increasingly exposed to time varying energy prices and operational variability, motivating digital twin models that can provide reliable, interpretable, and computationally tractable predictions in near real time. This work focuses on the empirical modeling core of the Empirical Model Based Optimization (EMBO) digital twin: adaptive L_1 -regularized regression for power forecasting from reverse osmosis (RO) flowrate setpoints. We construct a cubic polynomial feature library in permeate, concentrate, and recirculation flowrates and identify sparse regression models for power prediction. A key challenge arises during early operation or under limited data availability, where standard cross validation rules for selecting the regularization parameter (e.g., $\lambda_{\min}$ or λ_{1SE}) can lead to unstable sparsity patterns or degenerate models with all coefficients shrunk to zero. To address this, we propose a structured, λ selection strategy that explicitly controls model complexity while preventing degeneracy. A complementary 48-hour online digital-twin case study on the physical pilot demonstrates hourly retraining of the learned sparse power model on newly acquired pilot data, its integration within a 168-hour receding-horizon optimization problem, and the transfer of optimized permeate, concentrate, and recirculation flowrate setpoints to the pilot control interface. A representative optimization horizon generated after approximately two days of online operation demonstrates tariff-responsive load shifting, with reduced predicted power demand during high-tariff periods and increased operation during lower-tariff periods while maintaining the same cumulative production target. The results show that structured sparse regression provides an interpretable and computationally tractable foundation for digital twin decision support in water reuse operations.

Keywords. reverse osmosis, water reuse systems, adaptive L_1 regularization.

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1. Introduction

Advanced water reuse and desalination systems are increasingly operated under dynamic conditions driven by variable influent characteristics, stringent performance requirements, and fluctuations in electricity prices [1–3]. In particular, reverse osmosis (RO) typically dominates plant energy consumption, and operational degrees of freedom such as permeate, concentrate, and recirculation flowrates exert direct influence on aggregate power demand. Accurate and fast prediction of power consumption as a function of these controlled sensor variables is therefore a critical capability for energy aware operation and for tariff responsive planning [4–6].

Digital twins (DTs) provide a framework for continuously updated virtual models that enable monitoring, prediction, and decision support for physical assets [7]. A central challenge in DT development is the identification of virtual models that are both accurate and operationally tractable [2]. High-fidelity first-principles models can be difficult to calibrate and update frequently, especially under changing process conditions. Conversely, highly flexible purely data-driven models may achieve strong predictive accuracy, but often function as black-box predictors with limited interpretability and reduced transparency for operational decision-making [2, 8]. These limitations are particularly restrictive in real-time settings, where models must be retrained regularly, executed on conventional hardware, and remain reliable even when training data are limited.

Sparse empirical modeling offers a practical compromise between fidelity and tractability. By combining a physics-consistent feature library with regularized regression, sparse models can produce interpretable relationships between control variables and power demand, while limiting overfitting and enabling rapid retraining [9–11]. In this context, the selection of the least absolute shrinkage and selection operator (LASSO) regularization parameter λ is central: overly weak regularization may lead to unstable coefficient estimates, whereas overly strong regularization can collapse the model and yield zero coefficient (degenerate) models. This risk is amplified during early operation, when only a small number of training samples are available and cross-validation error estimates exhibit high variance.

This paper focuses on the empirical modeling component of the Empirical Model-Based Optimization (EMBO) digital twin [12], with an emphasis on adaptive L_1 penalization for robust power prediction from sensor-measured flowrate variables. We develop a structured λ -selection strategy that controls sparsity while preventing degeneracy, thereby ensuring that an interpretable model remains available across operating regimes. To address online engineering validation, we include a 48-hour online case study in which the sparse power model is embedded within a receding-horizon digital-twin loop and used to update RO flowrate setpoints on the physical pilot through the pilot control interface. The remainder of the paper is organized as follows. Section 2 introduces the pilot system. Section 3 presents the empirical modeling formulation, including the adaptive λ -selection logic, evaluates the impact of structured regularization on sparsity patterns and model stability, and reports the 48-hour digital-twin case study. Section 4 concludes the paper and discusses implications for longer-duration digital twin deployment.

2. Physical Pilot and Measurements

This section describes the experimental pilot platform used to generate training data. Figure 1(a) shows the pilot system used in this study, which replicates key operational characteristics of an advanced water reuse facility. The treatment train comprises three units in series: ultrafiltration (UF), reverse osmosis (RO), and ultraviolet/advanced oxidation (UV/AOP). Because RO is the primary energy consumer and provides direct operational degrees of freedom, this work focuses on the RO subsystem (Figure 1(b)).

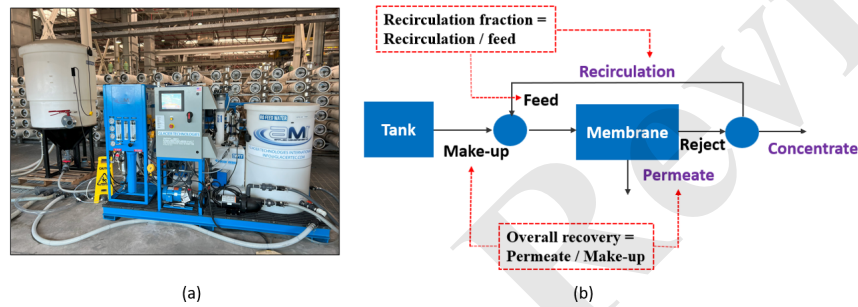


Figure 1. (a) UF-RO-UV/AOP pilot system used for data collection. (b) RO subsystem schematic highlighting the permeate, concentrate, and recirculation flowrate variables (in purple) used as inputs to the empirical model.

Three manipulated flowrate variables are considered: permeate, concentrate, and recirculation flowrates. These quantities determine internal RO flows (e.g., feed, reject, and make up streams) and therefore provide a minimal yet complete set of inputs for power prediction. The response variable is the aggregate electrical power demand of the pilot. A model-free exploratory analysis, as illustrated in figure 2, indicates strong positive associations between aggregate power demand and each flowrate variable, as well as the total flowrate (Pearson correlation ≈ 0.83). The corresponding Pearson correlation coefficients are 0.780 for permeate, 0.593 for recirculation, and 0.427 for concentrate flowrates. These observed relationships motivate the polynomial feature construction described next.

3. Digital Twin for Power Prediction & Results

A data-driven digital twin was developed in Python within the EMBO framework [12]. The primary methodological focus of the present work is the empirical power model that maps RO flowrate inputs to aggregate power demand. To evaluate this model under on-line operation, the empirical model was evaluated in a 48-hour online pilot case study in which the sparse power model was retrained hourly using newly acquired sensor data and embedded within a receding-horizon optimization loop. At each hourly cycle, the updated model was used to compute a 168-hour tariff-aware flowrate schedule for the permeate, concentrate, and recirculation streams; the first-hour setpoints from this schedule were then transferred to the pilot programmable logic controller/human-machine interface (PLC/HMI) for implementation. This two-day case study evaluates model avail-

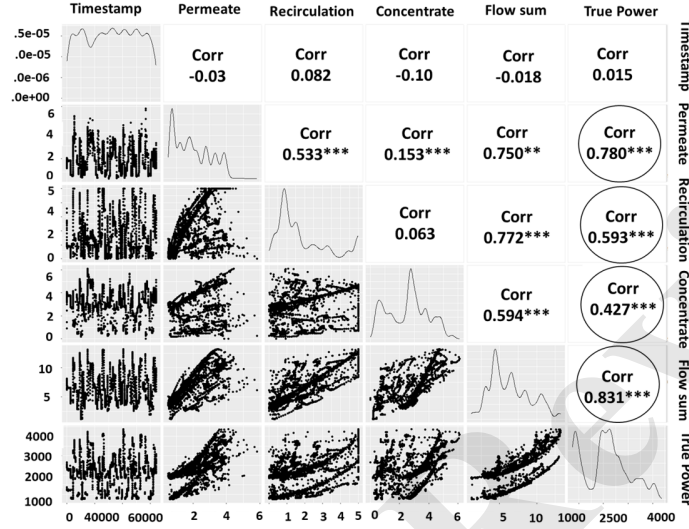


Figure 2. Pairwise scatter-plot matrix illustrating empirical relationships between aggregate power demand and flowrate variables. Pearson correlation coefficients demonstrate positive associations between power consumption and permeate, concentrate, and recirculation flowrates.

ability, computational feasibility within the one-hour decision interval, and bidirectional interaction between the digital twin and the physical pilot.

Let $\mathbf{X} \in \mathbb{R}^{n \times 3}$ denote the input matrix with columns corresponding to concentrate, permeate, and recirculation flowrates and n number of observations, and let $\mathbf{Y} \in \mathbb{R}^n$ denote the measured aggregate power demand. To capture nonlinearities and cross-coupling between flowrates, we construct a polynomial feature expansion up to degree 3. With $p = 3$ input variables and maximum degree $D = 3$ (cubic polynomial), the total number of monomials excluding the constant term is $\sum_{d=1}^D \binom{p+d-1}{d} = \binom{3}{1} + \binom{4}{2} + \binom{5}{3} = 3 + 6 + 10 = 19$. Thus, the feature library contains 3 first-order terms, 6 second-order terms, and 10 third-order terms, yielding 19 candidate terms. The resulting expansion can be written as $\Phi(\mathbf{X}) = [X_1, X_2, X_3, X_1^2, X_2^2, X_3^2, X_1X_2, X_1X_3, X_2X_3, \dots]$.

Because this feature library can overfit when n is small, we estimate regression coefficients via LASSO [9, 13]: $\hat{\beta} = \arg \min_{\beta} \{ \|\mathbf{Y} - \Phi(\mathbf{X})\beta\|_2^2 + \lambda \|\beta\|_1 \}$, where $\lambda > 0$ controls the sparsity accuracy trade off. Prior to model training, both the input features and the response variable (power) are standardized to zero mean and unit variance. Consequently, the regularization parameter λ is applied in the normalized feature space. All cross-validation errors reported in this study, including the mean squared error (MSE) values in Table 1, are computed in the standardized response space and are therefore dimensionless. Larger λ promotes stronger shrinkage and fewer active terms, improving interpretability but potentially degrading predictive performance if overly aggressive.

Model training uses 10-fold cross validation over a logarithmically spaced grid $\lambda \in [10^{-4}, 10^2]$. Let $\lambda_{\min}$ denote the value that minimizes the cross validated mean squared error (CV MSE), and let λ_{1SE} denote the largest λ whose CV MSE lies within one standard error of the minimum. While λ_{1SE} is often preferred for parsimony, in early

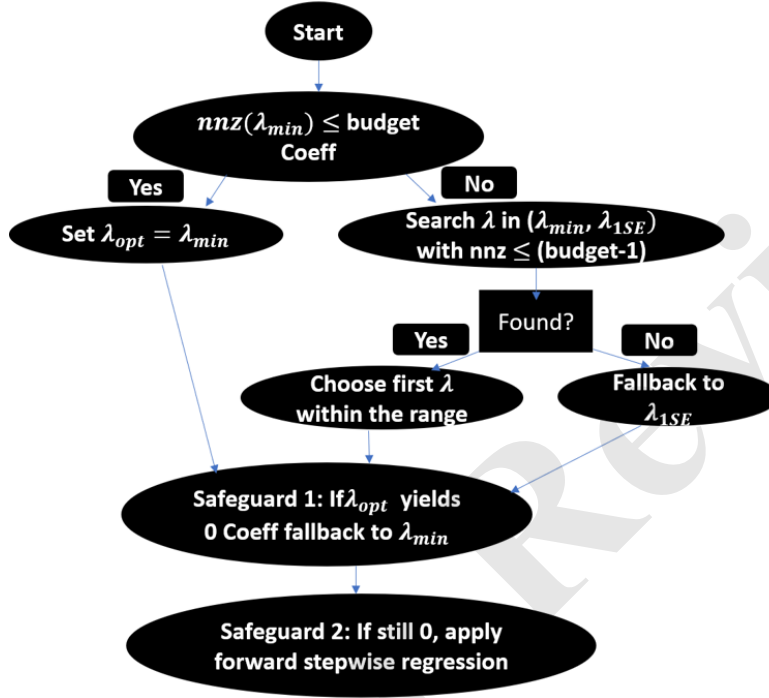


Figure 3. Structured selection and safeguard logic for adaptive L_1 regularization in the EMBO empirical model. Here nnz stands for number of nonzero coefficients.

operation the cross validation error bands can be wide, and λ_{1SE} may become overly conservative, yielding very sparse or degenerate models (all coefficients shrunk to zero). Such collapse is unacceptable in a real time digital twin because it removes the predictive mapping needed for decision support. Because the polynomial feature library contains 19 candidate terms, we impose an empirical sparsity budget to maintain interpretability and robustness, particularly under limited-data conditions.

To ensure that a nondegenerate and interpretable model is available throughout deployment, a structured selection rule is applied that explicitly controls model size and includes fail safe fallbacks:

1. **Primary rule:** select λ_{min} if the number of nonzero coefficients at λ_{min} is ≤ 8 (sparsity budget).
2. **Secondary rule:** otherwise scan from λ_{min} toward λ_{1SE} and select the first λ yielding ≤ 7 nonzero coefficients.
3. **Fallback rule:** if no such λ exists, set $\lambda_{opt} = \lambda_{1SE}$.
4. **Safeguard 1:** if $\lambda_{opt} = \lambda_{1SE}$ yields a degenerate model, revert to λ_{min} .
5. **Safeguard 2:** if λ_{min} also yields zero coefficients, fit a minimal linear model via forward stepwise selection on the original (non-polynomial) inputs.

The thresholds of 8 and 7 active coefficients are heuristic sparsity budgets chosen to retain a compact subset of dominant terms (approximately fewer than half of the candi-

date features), rather than strict constraints. If no value of λ in $[\lambda_{\min}, \lambda_{1SE}]$ satisfies the sparsity target, the procedure defaults to λ_{1SE} .

Figure 3 summarizes this decision logic. Figure 4 and Table 1 illustrate the motivation: with sufficient data (e.g., after twenty four hours), cross validation uncertainty is reduced and conventional rules yield stable sparse models, whereas with only two hours of data, structured selection avoids premature collapse while maintaining sparsity. A sensitivity analysis of the sparsity-budget thresholds is provided in Appendix A (Figure 8 and Table 3), and a retrospective comparison with alternative LASSO formulations is provided in Appendix B (Figures 9 and 10).

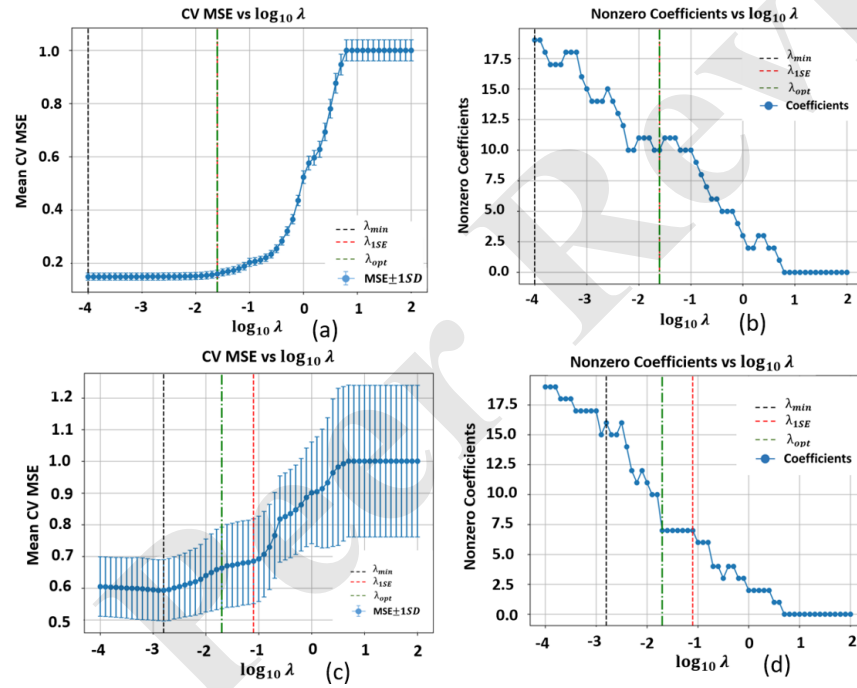


Figure 4. Cross-validation performance and sparsity behavior under different regularization levels: (a) Cross-validated mean squared error (MSE) versus λ (log scale) after 24 hours of operation, (b) Number of nonzero coefficients versus λ (log scale) after 24 hours of operation, (c) Cross-validated MSE versus λ (log scale) after 2 hours of operation, (d) Number of nonzero coefficients versus λ (log scale) after 2 hours of operation.

Table 1. Representative cross validation outcomes illustrating structured λ selection under limited (2 hours) and expanded (24 hours) training data. MSE values are reported in standardized (dimensionless) units.

Case	$\lambda_{\min}$	λ_{1SE}	λ_{opt}	MSE $\lambda_{\min}$	MSE λ_{opt}	Nonzeros
24 hours	0.000099	0.025119	0.025119	0.148914	0.159440	11
2 hours	0.0015849	0.0794328	0.019953	0.59325	0.685191	7

Next, for any candidate flowrate input matrix $\mathbf{X}_{\text{cand}} \in \mathbb{R}^{n \times 3}$ (n is number of observations), power is predicted as $\hat{\mathbf{Y}} = \Phi(\mathbf{X}_{\text{cand}})\hat{\beta}$. The sparse structure of $\hat{\beta}$ provides inter-

pretability by identifying the dominant flowrate terms and interactions that drive power demand. Figure 5 (a) reports feature importance for a representative model identified after twenty four hours, where 11 of the 19 candidate terms remain active (approximately 42.1% sparsity), and first order flowrate terms dominate the learned mapping. Figure 5(b) demonstrates close agreement between the predicted power (orange) and the experimentally measured power (green), indicating good predictive fidelity of the empirical model.

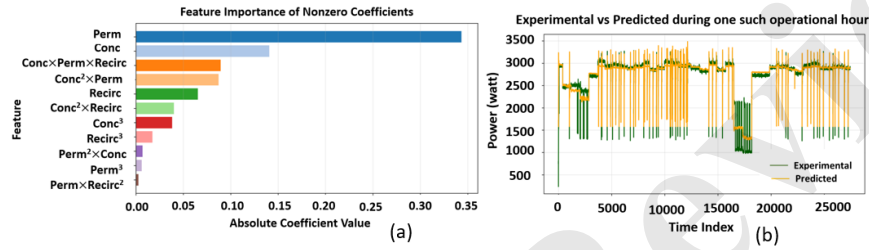


Figure 5. (a) Feature importance derived from the sparse model after 24 hours of operation where conc, perm and recirc stands for concentrate, permeate and recirculation flowrates respectively, (b) Comparison of predicted power (in orange) and experimental power (in green) during same period of operation.

3.1. Forty-Eight-Hour Online Digital Twin Case Study

To evaluate operation of the learned sparse power model within a real-time digital-twin workflow, the structured sparse model was embedded within the EMBO digital-twin loop and operated with the physical pilot for a 48-hour online case study. The case study was designed to assess whether model calibration, SEC minimization, and setpoint feedback could be executed repeatedly within the one-hour control interval using conventional computing resources and the pilot control interface. This experiment therefore evaluates the complete model-in-the-loop workflow: sensor data acquisition from the physical pilot, sparse model updating, tariff-aware optimization, and implementation of optimized flowrate setpoints on the pilot.

At each hourly control cycle, recent measurements of permeate, concentrate, and recirculation flowrates, together with aggregate pilot power demand, were transferred from the pilot data system to the digital twin. The cubic polynomial LASSO model was then retrained using the structured λ -selection rule described above. The updated sparse model was embedded in a receding-horizon optimization problem that computed a 168-hour sequence of candidate flowrate setpoints. Because three flowrate variables were optimized at each hourly step, the resulting decision vector contained $168 \times 3 = 504$ setpoint variables. The optimization was formulated to minimize the tariff-weighted specific energy cost (SEC) required to produce a user-defined cumulative volume of permeate over the 168-hour horizon.

Using the hourly electricity tariff vector $\mathbf{c} \in \mathbb{R}^{168}$, the tariff-weighted SEC objective over the 168-hour horizon can be written as

$$J_{\text{SEC}}(\mathbf{X}_{\text{cand}}) = \frac{\sum_{t=1}^{168} c_t \hat{Y}_t \Delta t}{V_{\text{perm,target}}},$$

where $\Delta t = 1$ hour and $V_{\text{perm,target}}$ is the prescribed cumulative permeate production target over the 168-hour optimization horizon. In the implementation, the same cumulative permeate production target was used for both optimized and nominal steady-state operation. Therefore, reductions in J_{SEC} reflect temporal load shifting in response to the tariff schedule rather than a reduction in water production.

After solving the optimization problem at each hourly update, only the first-hour setpoints for permeate, concentrate, and recirculation flowrates were transferred to the pilot PLC/HMI for implementation. At the next hourly cycle, newly acquired pilot data were incorporated, the sparse power model was updated, and a new 168-hour optimization horizon was computed. This receding-horizon implementation enables continuous adaptation while maintaining compatibility with the real-time control interval.

Figure 6 shows a representative 168-hour optimization horizon generated at the 25th hour of the 48-hour online case study. Figure 6 (a) shows the optimized trajectory to reduce the tariff-weighted SEC relative to nominal operation for the representative horizon by 38 %. The optimized operating schedule reduces predicted power demand during high-tariff periods and increases production during lower-tariff periods, while maintaining the same cumulative permeate production target as nominal steady-state operation (in Figure 6 (b)) .

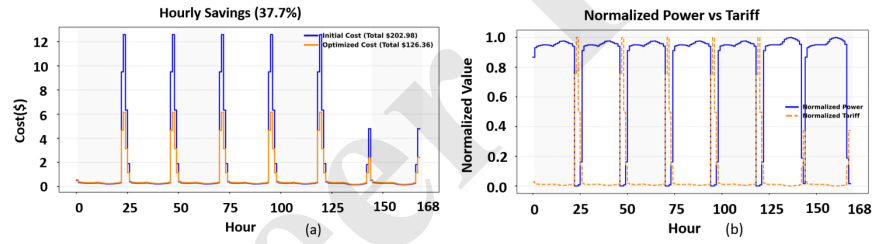


Figure 6. Representative 168-hour optimization horizon generated at the 25th hour of the 48-hour online digital-twin case study. (a) Comparison of hourly electricity cost under nominal steady-state operation and optimized digital-twin control, (b) Predicted power demand overlaid with the normalized electricity tariff, demonstrating tariff-aware load shifting. The optimized schedule reduces power demand during high-tariff periods and reallocates production to lower-tariff periods while maintaining the same cumulative permeate production target over the horizon.

Figure 7 summarizes the first-hour setpoints applied to the physical pilot during the full 48-hour case study. In contrast to Figure 6, which presents the complete 168-hour optimization horizon generated at one representative update, Figure 7 reports the sequence of control actions actually sent to the pilot at successive hourly updates. The applied setpoints follow the expected tariff-responsive structure, with reduced flowrates during high-tariff intervals and increased flowrates during lower-tariff intervals. This temporal pattern indicates consistent hourly setpoint generation under online receding-horizon operation.

Across the 48 hourly control cycles, the structured λ -selection rule maintained a nonzero sparse power model for use in the optimization step. This behavior was particularly important during early operation, when limited training data can lead to unstable cross-validation behavior and excessive coefficient shrinkage. By preventing fully sparse degenerate models, the proposed rule ensured that a usable empirical mapping between

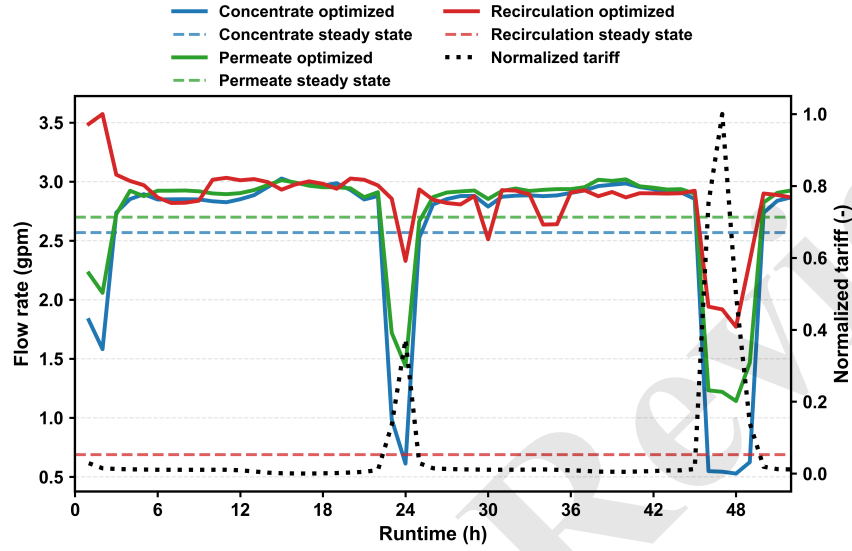


Figure 7. Hourly flowrate setpoints applied to the physical pilot during the 48-hour online digital-twin case study. Solid lines denote optimized setpoints implemented at each hourly update for concentrate (blue), permeate (green), and recirculation (red) flowrates, while dashed lines indicate the corresponding steady-state operating values. The normalized electricity tariff is shown for reference. The optimized setpoints decrease during high-tariff periods and increase during low-tariff periods, demonstrating reliable tariff-aware setpoint updates under continuous real-time digital-twin operation.

RO flowrates and aggregate power demand remained available throughout the online case study.

The computational time required at each update was also compatible with the one-hour control interval. The LASSO calibration step required a median time of 1.23 seconds (s), with a mean of 2.47 s and a range of 0.97–6.25 s. The subsequent constrained optimization step required a median time of 183.16 s, with a mean of 191.58 s and a range of 170.01–239.61 s. The total end-to-end computational time, including data loading, model calibration, and optimization, had a median of 186.81 s and a mean of 195.31 s. Thus, the complete digital-twin update required only a small fraction of the available one-hour interval, supporting online receding-horizon operation with substantial computational margin.

A summary of the 48-hour online digital-twin case study is provided in Table 2.

4. Conclusion and Future Work

This work presents a structured adaptive L_1 -regularized empirical modeling framework for power prediction in a water reuse pilot system. The proposed λ -selection strategy mitigates degeneracy and over-sparsification during early operation, while preserving model interpretability as training data accumulate. The resulting sparse power model provides a compact mapping from RO flowrate variables to aggregate power demand and is there-

Table 2. Summary of the 48-hour online digital-twin operation case study.

Quantity	Case-study value
Deployment duration	48 hours of online operation with the physical pilot
Control update interval	1 hour
Model update	Hourly retraining of the cubic LASSO power model using the structured λ -selection strategy
Prediction/optimization horizon	168 hours, corresponding to a 7-day tariff-aware receding horizon
Optimization variables	$168 \times 3 = 504$ hourly flowrate setpoints for permeate, concentrate, and recirculation streams
Optimization objective	Minimize tariff-weighted specific energy cost (SEC) for a fixed cumulative permeate production target
Implemented action	First-hour optimized flowrate setpoints transferred to the pilot PLC/HMI at each hourly update; a new 168-hour horizon was computed after the next data update
Representative horizon	168-hour optimization horizon generated at the 25th hour of operation, shown in Figure 6
Full case-study output	Hourly implemented setpoint sequence over 48 hours, shown in Figure 7
Comparison baseline	Nominal steady-state operation with the same cumulative permeate production target
Representative outcome	Approximately 38% lower tariff-weighted SEC relative to nominal steady-state operation for the representative horizon shown in Figure 6
LASSO calibration time	Median 1.23 s; mean 2.47 s; range 0.97–6.25 s
Constrained optimization time	Median 183.16 s; mean 191.58 s; range 170.01–239.61 s
End-to-end computational time	Median 186.81 s; mean 195.31 s, including data loading, model calibration, and optimization

fore suitable for use in real-time digital twin decision support. A 48-hour online case study demonstrates that the model can be embedded within a receding-horizon digital-twin loop, retrained hourly on pilot data, and used to generate flowrate setpoints that are transferred to the physical pilot control interface. The case study shows tariff-responsive operation, with reduced predicted power demand during high-tariff periods and compensating production during lower-tariff periods, while maintaining the same cumulative permeate production target. These results demonstrate the practical feasibility of using structured sparse regression as the empirical modeling core of a real-time digital twin for energy-aware operation of water reuse systems.

Future work will extend the online evaluation period beyond the 48-hour case study reported here to assess longer-duration robustness under broader operating conditions, including changes in influent characteristics, membrane condition, tariff structure, and equipment constraints. Longer evaluations will also be used to quantify the long-term effects of dynamic flowrate operation on membrane performance, actuator behavior, and water quality. As the developed model is empirical, it is intended for use within the physically allowable and data-supported operating region of the pilot system; extrapolation to substantially different feed-water composition, membrane condition, fouling state, equipment configuration, or operating regime requires additional validation, recalibration, or integration with physics-based process models. Future research will further

investigate hybrid digital twin frameworks that combine empirical models with physics-based simulators for full-scale water reuse facilities operating under human-in-the-loop supervisory control.

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Data Availability Statement

Data cannot be made publicly available; readers should contact the corresponding author for further details.

Conflict of Interest

The authors declare that they have no conflict of interest.

A. Supplementary Information: Sensitivity of Sparsity-Budget Thresholds

To evaluate whether the proposed structured λ -selection rule was sensitive to the empirical sparsity-budget thresholds, a sensitivity analysis was performed using the saved hourly training data from the closed-loop digital-twin deployment. The first 20 hourly training datasets were replayed, and the primary nonzero-coefficient budget was varied from 5 to 12, with the corresponding secondary budget set to one coefficient fewer than the primary budget. For each budget, the same polynomial feature library, standardization procedure, 10-fold cross-validation grid, and safeguard logic were retained; only the sparsity-budget threshold was changed. Model performance was evaluated using one-step-ahead prediction on the next unseen hour available from the following training data. Median error and interquartile ranges were used to summarize prediction performance because the earliest startup data can produce isolated high-error outliers. As shown in Figure 8 and Table 3, the deployed 8/7 budget retained a compact model with a median of 7 active coefficients and a median one-step-ahead MAPE of 2.27%. The neighboring budgets produced comparable median prediction errors, while higher budgets retained more active polynomial terms and reduced the cross-validation MSE penalty relative to $\lambda_{\min}$. These results indicate that the selected 8/7 rule provides a practical compromise between predictive accuracy and model parsimony, rather than being an isolated threshold choice.

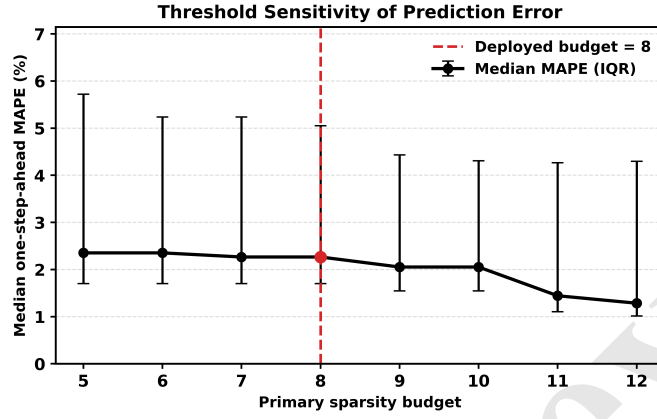


Figure 8. Sensitivity of one-step-ahead prediction error to the primary sparsity budget used in the structured λ -selection rule. Black markers denote the median MAPE across the evaluated early-operation data, error bars denote the interquartile range, and the red dashed line indicates the deployed primary budget of 8 active coefficients.

Table 3. Sensitivity of selected model size and prediction error to sparsity-budget thresholds. The primary/secondary budget pair denotes the maximum number of active coefficients used in the primary and secondary selection rules, respectively.

Primary/secondary budget	Median MAPE (%)	Median nonzero coefficients	Median CV MSE increase vs. $\lambda_{\min}$ (%)
5/4	2.35	7	12.77
6/5	2.35	7	11.27
7/6	2.27	7	10.91
8/7	2.27	7	10.61
9/8	2.05	8	9.20
10/9	2.05	9	9.17
11/10	1.44	10	5.90
12/11	1.28	11	3.12

B. Supplementary Information: Comparison with Alternative LASSO Formulations

To contextualize the proposed structured LASSO rule relative to established sparse-regression alternatives, a retrospective estimator comparison was performed using the same first 20 hourly training data from the closed-loop deployment. The comparison included the proposed structured LASSO, adaptive LASSO [14], degree-weighted LASSO [15], and Bayesian LASSO [16]. Adaptive LASSO was implemented as a two-stage weighted L_1 regression in which initial ridge-regression coefficients were used to assign coefficient-specific penalty weights, thereby reducing shrinkage on features with stronger preliminary support. Degree-weighted LASSO used the same polynomial feature library but applied larger penalty weights to higher-order polynomial terms, favoring simpler lower-order flowrate effects unless higher-order interactions were strongly supported by the data. Bayesian LASSO was implemented using a Laplace prior represented through a normal-exponential scale-mixture formulation and sampled using Gibbs sampling. Each method was trained on the same hourly data and evaluated using one-step-

ahead prediction of aggregate power demand on the next unseen hour from the following training file. Because the underlying pilot data were generated during the proposed closed-loop digital-twin deployment, this analysis should be interpreted as a retrospective estimator comparison on the observed data stream, not as an alternative closed-loop controller deployment. As shown in Figures 9 and 10, adaptive, degree-weighted, and Bayesian LASSO produced slightly lower median one-step-ahead MAPE values than the proposed structured rule, but retained substantially denser models. The proposed structured LASSO achieved a median MAPE of 2.27% with a median of 7 active coefficients, whereas adaptive and degree-weighted LASSO retained a median of 13 active coefficients and Bayesian LASSO retained a median of 17 active coefficients. Thus, the purpose of the proposed method is not solely to minimize retrospective prediction error, but to provide a stable, nondegenerate, and interpretable empirical model that can be recalibrated during hourly closed-loop digital-twin operation and can yield usable optimization results after the first few startup cycles.

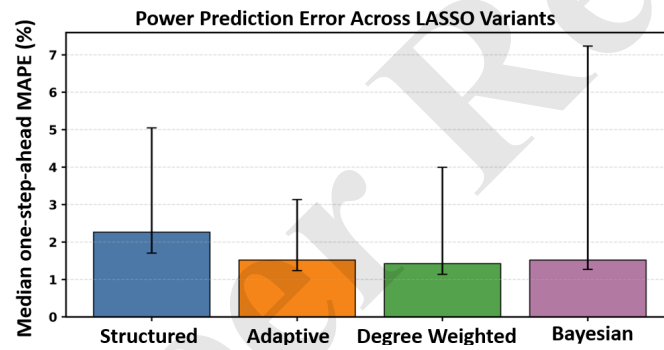


Figure 9. Retrospective comparison of one-step-ahead power-prediction error across LASSO variants using the same early-operation training data. Bars denote median MAPE and error bars denote the interquartile range.

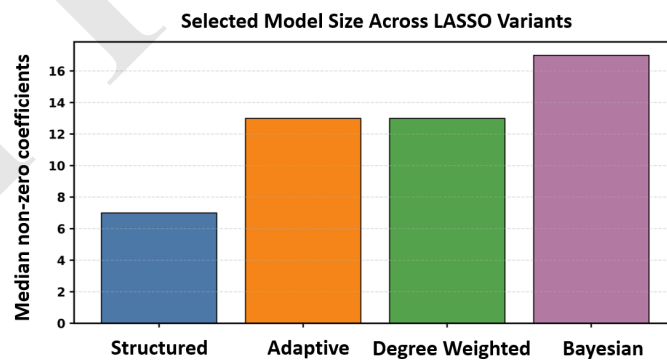


Figure 10. Retrospective comparison of selected model size across LASSO variants. Bars denote the median number of active polynomial coefficients selected across the evaluated early-operation data.

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Response to Reviewers

We would like to sincerely thank the Editor and the reviewers for their insightful and constructive comments. We have carefully revised the manuscript and believe that the revised version substantially improves the technical clarity, validation depth, and engineering relevance of the paper. We respond point-by-point to each reviewer comment below. The original reviewer comments are shown in red, our responses are provided in black, and additions or modifications in the revised manuscript are indicated in italics.

We respectfully note that this manuscript is intended for the ENDE2025 Special Issue, which follows a page-constrained format. However, in response to the reviewers' recommendations, the revised submission has been expanded to 11 pages, including an additional 3 pages of supplementary information. The major additions include:

- a new 48-hour online digital-twin case study demonstrating hourly model retraining, 168-hour receding-horizon optimization, and transfer of optimized setpoints to the pilot PLC/HMI;
- a supplementary sensitivity analysis of sparsity-budget thresholds;
- a supplementary comparison with Adaptive LASSO, Degree-weighted LASSO, and Bayesian LASSO;
- revisions to the abstract, introduction, discussion, conclusion, and future-work sections to reflect the added case study and clarify the scope and limitations of the proposed approach.

These revisions were made to directly address the reviewers' concerns regarding engineering validation, threshold sensitivity, comparison with alternative sparse-regression methods, and the practical relevance of the proposed structured L1-regularization framework.

Referee: 1

1. Although the authors justify the choice of LASSO by the need for sparsity, the manuscript still lacks a direct quantitative comparison with other regularization methods, such as Ridge Regression or Elastic Net. This is particularly important for scenarios involving a critically small number of data points.

We thank the reviewer for this thoughtful comment. We agree that comparisons among different regularization methods, including Ridge Regression and Elastic Net, can provide useful perspective in broader empirical modeling studies. However, the primary objective of the present manuscript is not to benchmark alternative regularization methods, but rather to address a specific stability issue that arises within the L_1 -regularized LASSO framework under limited-data conditions. The use of LASSO in this work is deliberate because sparsity and interpretability are central requirements for the intended digital twin application. LASSO promotes sparse models by shrinking some coefficients exactly to zero, thereby enabling automatic feature selection and identification of dominant flowrate terms and interactions. In contrast, Ridge Regression applies an L_2 penalty that shrinks coefficients toward zero but can retain all candidate features in the model. As a result, Ridge Regression does not provide the same sparse feature-selection capability and does not encounter the same all-zero degeneracy issue addressed in this paper. Elastic Net combines L_1 and L_2 penalties and may be useful in other settings; however, it introduces an additional mixing parameter and does not directly target the specific problem

considered here, namely unstable sparsity selection and fully degenerate LASSO models during early operation with limited training data.

The contribution of this manuscript is therefore centered on a structured λ -selection and safeguard strategy that prevents degenerating sparse models while retaining interpretability and computational tractability. This issue is specific to sparse L1-regularized modeling and is not encountered in the same form in non-sparse Ridge Regression. We have clarified this rationale in the revised manuscript and have focused the limited manuscript space on validating the proposed LASSO-specific framework.

We added a new supplementary subsection titled “*Supplementary Information: Comparison with Alternative LASSO Formulations*,” in which the proposed structured LASSO rule is compared against Adaptive LASSO, Degree-weighted LASSO, and Bayesian LASSO using the same early-operation training data. We also added supplementary figures comparing one-step-ahead MAPE and the number of selected active coefficients across LASSO variants.

We also note that the manuscript is submitted to the ENDE2025 Special Issue, which enforces a strict page limit. For this reason, a comprehensive benchmarking study across multiple regularization frameworks has been left for future work.

In addition, the revised manuscript now includes a 48-hour online digital twin case study, which demonstrates that the proposed structured LASSO model remains usable during repeated hourly retraining under evolving online data availability (*Subsection: Forty-Eight-Hour Online Digital Twin Case Study*). This added case study supports the practical relevance of the proposed LASSO-specific stabilization strategy.

2. The selection of thresholds for the number of nonzero coefficients (e.g., ≤ 8 or 7) is based on empirical “complexity budgets.” It is recommended to include a sensitivity analysis of the results relative to changes in these thresholds.

We thank the reviewer for this helpful suggestion. We agree that the sparsity-budget thresholds used in the proposed structured λ -selection rule are empirical design choices and should be evaluated for sensitivity. In response, we have added a new supplementary subsection titled “*Supplementary Information: Sensitivity of Sparsity-Budget Thresholds*” to the revised manuscript.

In this analysis, we replayed the first 20 hourly training data from the closed-loop digital-twin deployment and varied the primary nonzero-coefficient budget from 5 to 12. For each primary budget, the corresponding secondary budget was set to one coefficient fewer than the primary budget, consistent with the structure of the proposed selection rule. All other modeling components were kept fixed, including the polynomial feature library, standardization procedure, 10-fold cross-validation grid, and safeguard logic. Therefore, the analysis isolates the effect of the sparsity-budget threshold itself.

Model performance was evaluated using one-step-ahead prediction of aggregate power demand on the next unseen hourly data. Median error and interquartile ranges were used to summarize performance, since the earliest startup data can produce isolated high-error outliers under limited-data conditions. The results show that the deployed 8/7 sparsity-budget rule achieved a median one-step-ahead MAPE of 2.27% with a median of 7 active coefficients. Neighboring sparsity budgets produced comparable median prediction errors, while larger budgets retained more active polynomial terms and reduced the parsimony advantage of the proposed structured rule.

These results indicate that the selected 8/7 threshold is not an isolated or uniquely tuned choice. Rather, it provides a practical compromise between predictive accuracy and model parsimony. The sensitivity analysis therefore supports the interpretation of the proposed thresholds as empirical complexity budgets designed to maintain a compact, interpretable, and nondegenerate sparse model during early online operation, rather than as theoretically optimal threshold values.

Changes made in the revised manuscript

We added a supplementary subsection titled “*Supplementary Information: Sensitivity of Sparsity-Budget Thresholds*.” In this section, the primary sparsity-budget threshold was varied from 5 to 12 active coefficients, with the secondary threshold set to one coefficient fewer than the primary threshold.

To evaluate whether the proposed structured λ -selection rule was sensitive to the empirical sparsity-budget thresholds, a sensitivity analysis was performed using the saved hourly training data from the closed-loop digital-twin deployment. The first 20 hourly training datasets were replayed, and the primary nonzero-coefficient budget was varied from 5 to 12, with the corresponding secondary budget set to one coefficient fewer than the primary budget. For each budget, the same polynomial feature library, standardization procedure, 10-fold cross-validation grid, and safeguard logic were retained; only the sparsity-budget threshold was changed. Model performance was evaluated using one-step-ahead prediction on the next unseen hour available from the following training data. Median error and interquartile ranges were used to summarize prediction performance because the earliest startup data can produce isolated high-error outliers. As shown in Figure 1 and Table 1 (here), the deployed 8/7 budget retained a compact model with a median of 7 active coefficients and a median one-step-ahead MAPE of 2.27%. The neighboring budgets produced comparable median prediction errors, while higher budgets retained more active polynomial terms and reduced the cross-validation MSE penalty relative to λ_{min} . These results indicate that the selected 8/7 rule provides a practical compromise between predictive accuracy and model parsimony, rather than being an isolated threshold choice.

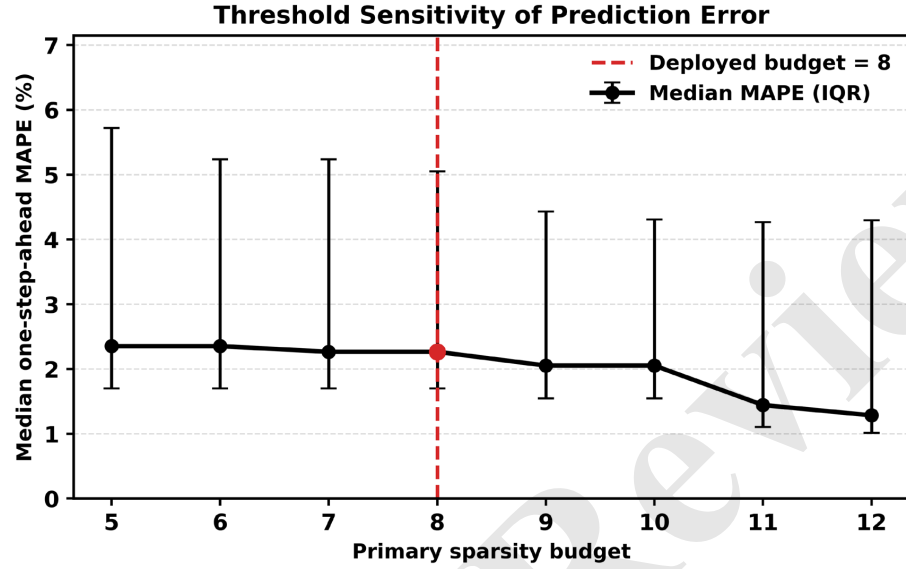


Figure 1: Sensitivity of one-step-ahead prediction error to the primary sparsity budget used in the structured lambda-selection rule. Black markers denote the median MAPE across the evaluated early-operation data, error bars denote the interquartile range, and the red dashed line indicates the deployed primary budget of 8 active coefficients.

Table 1: Sensitivity of selected model size and prediction error to sparsity-budget thresholds. The primary/secondary budget pair denotes the maximum number of active coefficients used in the primary and secondary selection rules, respectively.

Primary/secondary budget	Median MAPE (%)	Median nonzero coefficients	Median CV increase vs. λ_{min} (%)	MSE
5/4	2.35	7	12.77	
6/5	2.35	7	11.27	
7/6	2.27	7	10.91	
8/7	2.27	7	10.61	
9/8	2.05	8	9.20	
10/9	2.05	9	9.17	
11/10	1.44	10	5.90	
12/11	1.28	11	3.12	

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3. As the model is purely empirical, it does not incorporate the physical laws governing the reverse osmosis process. The authors should more clearly define the boundaries of its applicability, especially regarding extrapolation beyond the range of the training data.

We thank the reviewer for this important comment. All digital-twin models, including physics-based simulators, must respect the pilot's operating constraints, the present empirical model additionally depends on the range of conditions represented in the training data.

Therefore, the model should not be interpreted as a general extrapolative RO process model. Substantially different feed-water composition, salinity, fouling state, membrane condition, equipment configuration, or operating regime would require additional data, recalibration, and validation. Gradual process changes may be captured through hourly retraining when reflected in the measured data, but fast physicochemical changes not represented in the input variables remain challenging for a purely empirical model.

To address this limitation, we have revised the Conclusion and Future Work section to state that future work will investigate hybrid digital-twin frameworks combining empirical models with physics-based simulators, particularly for full-scale systems where extrapolation, water-quality dynamics, and fouling evolution will be considered.

Changes made in the revised manuscript:

As the developed model is empirical, it is intended for use within the physically allowable and data-supported operating region of the pilot system; extrapolation to substantially different feed-water composition, membrane condition, fouling state, equipment configuration, or operating regime requires additional validation, recalibration, or integration with physics-based process models. Future research will further investigate hybrid digital twin frameworks that combine empirical models with physics-based simulators for full-scale water reuse facilities operating under human-in-the-loop supervisory control.

4. In Table 1, the units of measurement for MSE must be explicitly stated. Although the authors mention that the values are dimensionless due to standardization, this should be reflected directly in the table caption or headers.

We clarified the reporting of MSE values in Table 1 by explicitly stating in the table caption that the MSE values are reported in standardized, dimensionless units.

5. Attention should be paid to the consistency of figure captions and formatting (e.g., between Fig. 4 and Fig. 5) to ensure compliance with publication standards.

We thank the reviewer for this helpful suggestion. We have reviewed the figures and ensured that all figure captions and formatting are consistent throughout the revised manuscript.

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6. The updated data availability statement ("Data cannot be made publicly available...") appears justified given the involvement of national laboratories. However, the authors should ensure this aligns strictly with the specific journal policy.

We thank the reviewer for this comment. We have verified that the revised Data Availability Statement is fully consistent with the journal's data-sharing and availability policies. We also note that the journal has published articles with varying approaches to data availability reporting. To ensure transparency and compliance, we have retained the explicit Data Availability Statement in the revised manuscript.

7. The authors should verify the order of the reference list to ensure citations follow the sequence of their appearance in the text (as previous versions appeared to use alphabetical order).

We thank the reviewer for this helpful comment. We have carefully reviewed the reference list and verified that, in the revised manuscript, all references are now ordered according to the sequence of their first appearance in the text.

Referee: 2

This paper focuses on the energy consumption prediction requirement of the reverse osmosis (RO) process in water reuse systems. Aiming at the degeneracy of LASSO models caused by limited sample data in digital twin scenarios, the authors propose a structured L1 regularization modeling framework with a heuristic safeguard mechanism. The topic aligns with the engineering requirements of energy-efficient operation and digital twin implementation in the water treatment industry, and has a basic application background. However, on the whole, the manuscript has multiple fundamental flaws that cannot be remedied through routine revisions, with a severe deficiency in core innovation. The main comments are as follows:

1. The core innovation of this paper is merely a set of heuristic rules for λ selection based on empirical thresholds. The authors fail to propose any innovative method with solid theoretical support for the core issue of LASSO regularization parameter selection. Furthermore, the manuscript does not clarify the specific substantial academic increment and technological breakthroughs of this work compared with well-established improved LASSO methods for small sample scenarios, including Adaptive LASSO, Bayesian LASSO, and Weighted LASSO, in either the Introduction or Discussion sections.

We thank the reviewer for this important and constructive comment. We agree that the original manuscript did not sufficiently contextualize the proposed structured L1-regularization rule relative to established sparse-regression alternatives such as Adaptive LASSO, Bayesian LASSO, and Weighted LASSO formulations. We also agree that the proposed approach should not be presented as a general theoretical solution to the broader problem of optimal LASSO regularization parameter selection.

To address this concern, we have revised the manuscript to more clearly state the intended scope of the contribution. The purpose of the proposed method is not to replace theoretically grounded LASSO parameter-selection frameworks, but to provide a practical and deployment-

oriented sparse-model selection strategy for online digital twin operation under limited-data conditions. In this setting, the empirical model must satisfy several simultaneous requirements: it must remain nondegenerate, be sufficiently sparse for interpretability, be recalibrated quickly during hourly updates, and remain usable inside a downstream receding-horizon optimization problem. The structured λ -selection and safeguard logic was therefore designed to address an operational failure mode observed during early-stage digital twin operation, namely excessive shrinkage leading to fully degenerate all-zero sparse models. The revised manuscript now clarifies that the active-coefficient thresholds are heuristic sparsity budgets selected to maintain compactness and avoid degeneracy rather than theoretically optimal regularization parameters.

In response to the reviewer's suggestion for comparison with improved LASSO methods, we added a new supplementary subsection titled "Supplementary Information: Comparison with Alternative LASSO Formulations." In this supplementary analysis, the proposed structured LASSO rule was compared retrospectively with Adaptive LASSO, Degree-weighted LASSO, and Bayesian LASSO using the same first 20 hourly training datasets from the closed-loop deployment. Each method was trained on the same hourly data stream and evaluated using one-step-ahead prediction of aggregate power demand for the next unseen hour. This comparison was designed to assess whether the proposed rule provides a useful balance between prediction accuracy and sparse model complexity under early-operation, limited-data conditions.

The supplementary comparison showed that Adaptive LASSO, Degree-weighted LASSO, and Bayesian LASSO produced slightly lower median one-step-ahead MAPE values than the proposed structured LASSO rule. However, these methods also retained substantially denser models. The structured LASSO achieved a median MAPE of 2.27% with a median of 7 active coefficients, whereas Adaptive LASSO and Degree-weighted LASSO retained a median of 13 active coefficients, and Bayesian LASSO retained a median of 17 active coefficients. Thus, the proposed method is not claimed to be the most accurate retrospective estimator in isolation. Rather, its practical value lies in providing a stable, nondegenerate, and interpretable sparse model that remains suitable for repeated online recalibration and downstream optimization during digital twin operation.

We have also revised the manuscript to clarify the technological contribution of the work. The proposed structured sparse model was not evaluated only as an offline estimator; it was embedded within a 48-hour online digital twin case study on the physical pilot system. In this case study, the sparse power model was retrained hourly using newly acquired pilot data, embedded into a 168-hour tariff-aware receding-horizon optimization problem, and used to generate first-hour flowrate setpoints that were transferred to the pilot PLC/HMI for implementation. This provides an engineering demonstration of the proposed sparse-modeling strategy in a model-in-the-loop digital twin workflow, rather than only a retrospective regression exercise.

Changes made in the revised manuscript

We added a new supplementary subsection titled “Supplementary Information: Comparison with Alternative LASSO Formulations,” in which the proposed structured LASSO rule is compared against Adaptive LASSO, Degree-weighted LASSO, and Bayesian LASSO using the same early-operation training data. We also added supplementary figures comparing one-step-ahead MAPE and the number of selected active coefficients across LASSO variants.

To contextualize the proposed structured LASSO rule relative to established sparse-regression alternatives, a retrospective estimator comparison was performed using the same first 20 hourly training data from the closed-loop deployment. The comparison included the proposed structured LASSO, adaptive LASSO [1], degree-weighted LASSO [2], and Bayesian LASSO [3]. Adaptive LASSO was implemented as a two-stage weighted L_1 regression in which initial ridge-regression coefficients were used to assign coefficient-specific penalty weights, thereby reducing shrinkage on features with stronger preliminary support. Degree-weighted LASSO followed the general weighted/reweighted L_1 penalization concept [2] and used the same polynomial feature library but applied larger penalty weights to higher-order polynomial terms, favoring simpler lower-order flowrate effects unless higher-order interactions were strongly supported by the data. Bayesian LASSO was implemented using a Laplace prior represented through a normal-exponential scale-mixture formulation and sampled using Gibbs sampling [3]. Each method was trained on the same hourly data and evaluated using one-step-ahead prediction of aggregate power demand on the next unseen hour from the following training file. Because the underlying pilot data were generated during the proposed closed-loop digital-twin deployment, this analysis should be interpreted as a retrospective estimator comparison on the observed data stream, not as an alternative closed-loop controller deployment. As shown in Figures 2 and 3 (here), adaptive, degree-weighted, and Bayesian LASSO produced slightly lower median one-step-ahead MAPE values than the proposed structured rule but retained substantially denser models. The proposed structured LASSO achieved a median MAPE of 2.27% with a median of 7 active coefficients, whereas adaptive and degree-weighted LASSO retained a median of 13 active coefficients and Bayesian LASSO retained a median of 17 active coefficients. Thus, the purpose of the proposed method is not solely to minimize retrospective prediction error, but to provide a stable, nondegenerate, and interpretable empirical model that can be recalibrated during hourly closed-loop digital-twin operation and can yield usable optimization results after the first few startup cycles.

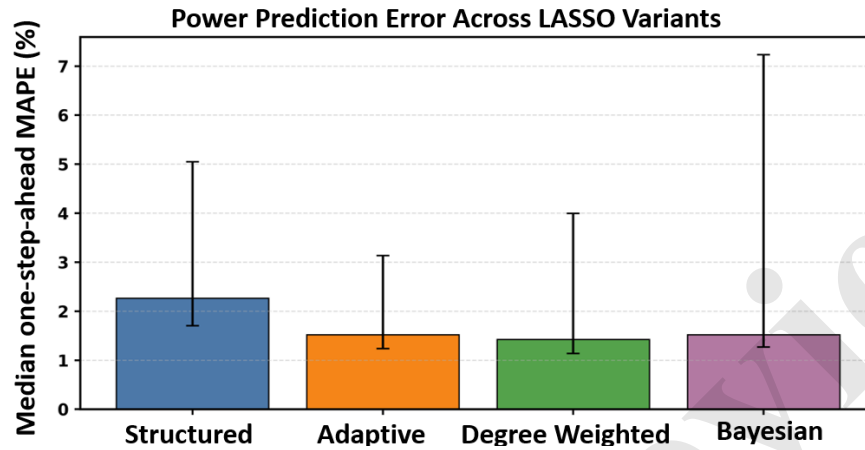


Figure 2: Retrospective comparison of one-step-ahead power-prediction error across LASSO variants using the same early-operation training data. Bars denote median MAPE and error bars denote the interquartile range.

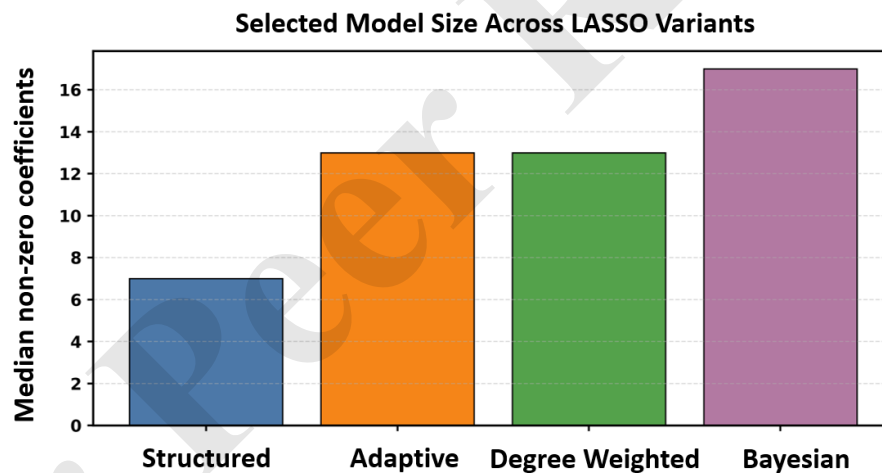


Figure 3: Retrospective comparison of selected model size across LASSO variants. Bars denote the median number of active polynomial coefficients selected across the evaluated early-operation data.

2. The tests are only completed on the dataset from a single pilot-scale system, without validation of robustness and generalization performance under multiple operating conditions, multiple data sources, and different noise levels.

We thank the reviewer for raising this important point. We agree that validation across multiple pilot systems, operating regimes, data sources, and noise conditions would provide a broader assessment of generalization performance. The present study is intentionally focused on a single physical water reuse pilot system because the objective is to evaluate whether the proposed structured sparse modeling framework can remain stable, interpretable, and computationally tractable within a real digital twin workflow. It is to be noted that this is one of the first applications of digital twin in water space industry.

To strengthen the engineering validation beyond the original offline analysis, we have added a new 48-hour online digital twin case study in the revised manuscript. In this case study, the sparse power model is retrained hourly using newly acquired pilot data, embedded within a 168-hour receding-horizon optimization problem, and used to generate flowrate setpoints that are transferred to the physical pilot control interface. This addition evaluates the model under evolving online data availability and changing operating setpoints, rather than only on a static historical dataset. Across the 48 hourly control cycles, the structured λ -selection rule maintained a nonzero sparse model for use in the optimization step, which is particularly important during early operation when limited training data can lead to unstable cross-validation behavior and excessive coefficient shrinkage.

We acknowledge that the study remains focused on one pilot-scale system. We have therefore revised the conclusion and future-work discussion to explicitly state that longer-duration evaluations across broader operating conditions are needed. These include changes in influent characteristics, membrane condition, tariff structure, equipment constraints, and additional water reuse systems. Given the strict page limitation of the special issue, we believe the added 48-hour online case study provides an appropriate and feasible extension within the present manuscript, while multi-site and multi-source validation is identified as an important direction for future work.

Changes made in the revised manuscript:

We added a new subsection titled “Forty-Eight-Hour Online Digital Twin Case Study” and revised the conclusion to acknowledge the need for longer-duration and broader validation under operating conditions, data sources, and system configurations.

To evaluate operation of the learned sparse power model within a real-time digital-twin workflow, the structured sparse model was embedded within the EMBO digital-twin loop and operated with the physical pilot for a 48-hour online case study. The case study was designed to assess whether model calibration, SEC minimization, and setpoint feedback could be executed repeatedly within the one-hour control interval using conventional computing resources and the pilot control interface. This experiment therefore evaluates the complete model-in-the-loop workflow: sensor data acquisition from the physical pilot, sparse model updating, tariff-aware optimization, and implementation of optimized flowrate setpoints on the pilot.

At each hourly control cycle, recent measurements of permeate, concentrate, and recirculation flowrates, together with aggregate pilot power demand, were transferred from the pilot data system to the digital twin. The cubic polynomial LASSO model was then retrained using the

structured lambda-selection rule described above. The updated sparse model was embedded in a receding-horizon optimization problem that computed a 168-hour sequence of candidate flowrate setpoints. Because three flowrate variables were optimized at each hourly step, the resulting decision vector contained $168 * 3 = 504$ setpoint variables. The optimization was formulated to minimize the tariff-weighted specific energy cost (SEC) required to produce a user-defined cumulative volume of permeate over the 168-hour horizon.

Using the hourly electricity tariff vector cinR^{168} , the tariff-weighted SEC objective over the 168-hour horizon can be written as $J_{\text{SEC}}(X_{\text{cand}}) = \sum_{t=1}^{168} c_t \hat{Y}_t \Delta t / V_{\text{perm,target}}$, where $\Delta t = 1\text{hr}$ and $V_{\text{perm,target}}$ is the prescribed cumulative permeate production target over the 168-hour optimization horizon. In the implementation, the same cumulative permeate production target was used for both optimized and nominal steady-state operation. Therefore, reductions in J_{SEC} reflect temporal load shifting in response to the tariff schedule rather than a reduction in water production.

After solving the optimization problem at each hourly update, only the first-hour setpoints for permeate, concentrate, and recirculation flowrates were transferred to the pilot PLC/HMI for implementation. At the next hourly cycle, newly acquired pilot data were incorporated, the sparse power model was updated, and a new 168-hour optimization horizon was computed. This receding-horizon implementation enables continuous adaptation while maintaining compatibility with the real-time control interval.

Figure 4 (here) shows a representative 168-hour optimization horizon generated at the 25th hour of the 48-hour online case study. Figure 4(a) shows the optimized trajectory to reduce the tariff-weighted SEC relative to nominal operation for the representative horizon by 38%. The optimized operating schedule reduces predicted power demand during high-tariff periods and increases production during lower-tariff periods, while maintaining the same cumulative permeate production target as nominal steady-state operation (in Figure 4(b)).

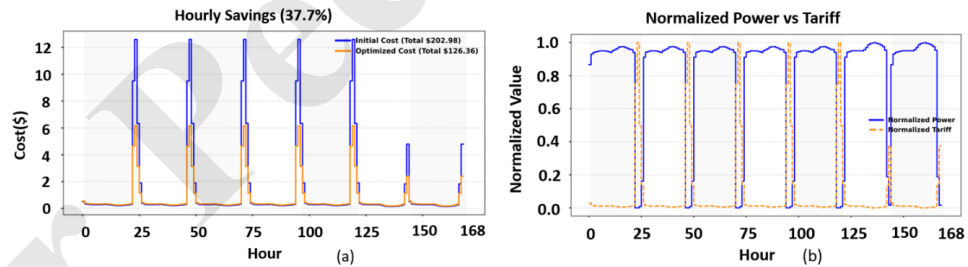


Figure 4: Representative 168-hour optimization horizon generated at the 25th hour of the 48-hour online digital-twin case study. (a) Comparison of hourly electricity cost under nominal steady-state operation and optimized digital-twin control. (b) Predicted power demand overlaid with the normalized electricity tariff, demonstrating tariff-aware load shifting. The optimized schedule reduces power demand during high-tariff periods and reallocates production to lower-tariff periods while maintaining the same cumulative permeate production target over the horizon.

Figure 5 (here) summarizes the first-hour setpoints applied to the physical pilot during the full 48-hour case study. In contrast to Figure 4, which presents the complete 168-hour optimization

horizon generated at one representative update, Figure 5 reports the sequence of control actions actually sent to the pilot at successive hourly updates. The applied setpoints follow the expected tariff-responsive structure, with reduced flowrates during high-tariff intervals and increased flowrates during lower-tariff intervals. This temporal pattern indicates consistent hourly setpoint generation under online receding-horizon operation.

Hourly flowrate setpoints applied to the physical pilot during the 48-hour online digital-twin case study. Solid lines denote optimized setpoints implemented at each hourly update for concentrate (blue), permeate (green), and recirculation (red) flowrates, while dashed lines indicate the corresponding steady-state operating values. The normalized electricity tariff is shown for reference. The optimized setpoints decrease during high-tariff periods and increase during low-tariff periods, demonstrating reliable tariff-aware setpoint updates under continuous real-time digital-twin operation.

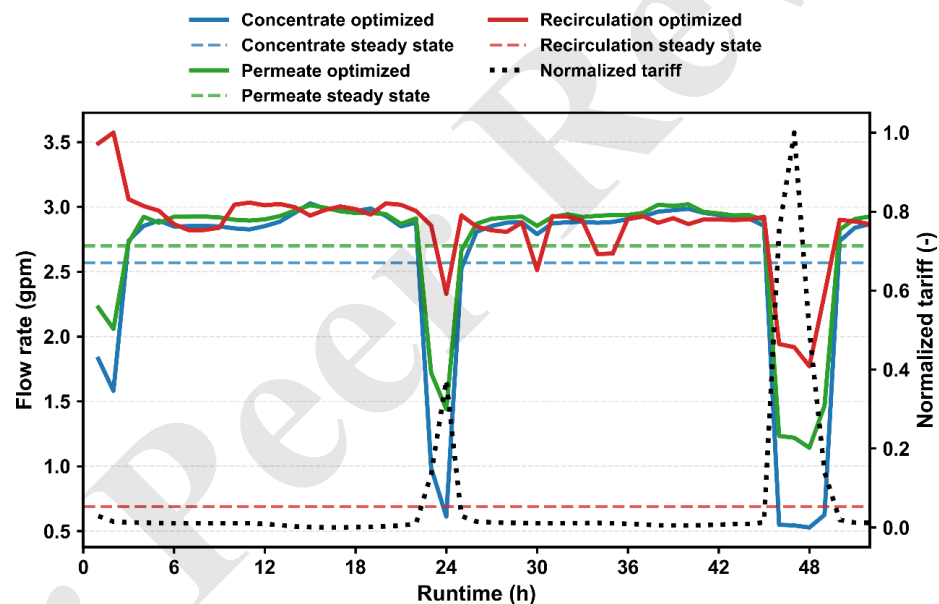


Figure 5: Hourly flowrate setpoints applied to the physical pilot during the 48-hour online digital-twin case study. Solid lines denote optimized setpoints implemented at each hourly update for concentrate (blue), permeate (green), and recirculation (red) flowrates, while dashed lines indicate the corresponding steady-state operating values. The normalized electricity tariff is shown for reference. The optimized setpoints decrease during high-tariff periods and increase during low-tariff periods, demonstrating reliable tariff-aware setpoint updates under continuous real-time digital-twin operation.

Across the 48 hourly control cycles, the structured lambda-selection rule maintained a nonzero sparse power model for use in the optimization step. This behavior was particularly important during early operation, when limited training data can lead to unstable cross-validation behavior

and excessive coefficient shrinkage. By preventing fully sparse degenerate models, the proposed rule ensured that a usable empirical mapping between RO flowrates and aggregate power demand remained available throughout the online case study.

The computational time required at each update was also compatible with the one-hour control interval. The LASSO calibration step required a median time of 1.23 seconds (s), with a mean of 2.47 s and a range of 0.97-6.25 s. The subsequent constrained optimization step required a median time of 183.16 s, with a mean of 191.58 s and a range of 170.01-239.61 s. The total end-to-end computational time, including data loading, model calibration, and optimization, had a median of 186.81 s and a mean of 195.31 s. Thus, the complete digital-twin update required only a small fraction of the available one-hour interval, supporting online receding-horizon operation with substantial computational margin.

A summary of the 48-hour online digital-twin case study is provided in Table 2.

Quantity	Case-study value
Deployment duration	48 hours of online operation with the physical pilot
Control update interval	1 hour
Model update	Hourly retraining of the cubic LASSO power model using the structured lambda-selection strategy
Prediction/optimization horizon	168 hours, corresponding to a 7-day tariff-aware receding horizon
Optimization variables	168 x 3 = 504 hourly flowrate setpoints for permeate, concentrate and recirculation streams
Optimization objective	Minimize tariff-weighted specific energy cost (SEC) for a fixed cumulative permeate production target
Implemented action	First-hour optimized flowrate setpoints transferred to the pilot PLC/HMI at each hourly update; a new 168-hour horizon was computed after the next data update
Representative horizon	168-hour optimization horizon generated at the 25th hour of operation, shown in Figure 4
Full case-study output	Hourly implemented setpoint sequence over 48 hours, shown in Figure 5
Comparison baseline	Nominal steady-state operation with the same cumulative permeate production target
Representative outcome	Approximately 38% lower tariff-weighted SEC relative to nominal steady-state operation for the representative horizon shown in Figure 4

LASSO calibration time	Median 1.23 s; mean 2.47 s; range 0.97--6.25 s
Constrained optimization time	Median 183.16 s; mean 191.58 s; range 170.01--239.61 s
End-to-end computational time	Median 186.81 s; mean 195.31 s, including data loading, model calibration, and optimization
Deployment duration	48 hours of online operation with the physical pilot

3. The manuscript only estimates the time consumption of model retraining, and fails to complete the core engineering validation in digital twin scenarios: specifically, the actual deployment and real-time operation tests of the model on mainstream industrial control hardware widely used in water treatment plants are not carried out.

We thank the reviewer for this constructive comment. We agree that digital-twin methods should be evaluated not only through offline computational timing estimates, but also through model-in-the-loop operation with physical systems and control interfaces. In response to this comment, we have substantially revised the manuscript by adding a 48-hour online digital-twin case study conducted on the physical water reuse pilot.

In the added case study, the structured sparse power model was embedded within the EMBO digital-twin loop and operated with the physical pilot over a 48-hour period. At each hourly control cycle, recent measurements of permeate, concentrate, and recirculation flowrates, together with aggregate pilot power demand, were transferred from the pilot data system to the digital twin. The cubic polynomial LASSO model was then retrained using the structured λ -selection strategy. The updated sparse model was subsequently embedded within a 168-hour receding-horizon optimization problem containing $168 \times 3 = 504$ hourly flowrate setpoint variables. After each optimization, the first hour optimized setpoints for permeate, concentrate and recirculation flowrates were transferred to the pilot PLC/HMI for implementation.

This revision directly addresses the reviewer's concern by moving beyond an offline estimate of retraining time. The revised manuscript now reports measured component-wise computational timing for the complete digital-twin update cycle, including data loading, model calibration, and constrained optimization. Specifically, across the 48 hourly update cycles, the LASSO calibration step required a median time of 1.23 s, with a mean of 2.47 s and a range of 0.97--6.25 s. The constrained optimization step required a median time of 183.16 s, with a mean of 191.58 s and a range of 170.01--239.61 s. The total end-to-end computational time, including data loading, model calibration, and optimization, had a median of 186.81 s and a mean of 195.31 s. Thus, the complete digital-twin update required only a small fraction of the available one-hour control interval, demonstrating substantial computational margin for online receding-horizon operation.

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8 The revised manuscript now demonstrates actual model-in-the-loop operation with a physical
9 pilot and its PLC/HMI control interface. This provides a meaningful engineering validation step
10 within the scope of the present study. We also note that the manuscript is submitted to the
11 ENDE2025 Special Issue, which enforces a strict page limit; therefore, full-scale industrial
12 deployment and longer-duration operation are identified as important extensions for future
13 work. The revised conclusion has been updated accordingly to state that future work will include
14 longer-duration evaluations under broader operating conditions, including full-scale water reuse
15 facilities and human-in-the-loop supervisory control.
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18 **Changes made in the revised manuscript:**

19 We added a new 48-hour online digital-twin case study, including hourly model retraining, 168-
20 hour receding-horizon optimization, transfer of first hour optimized flowrate setpoints to the
21 pilot PLC/HMI, and measured component-wise computational timing results. The revised
22 manuscript now reports the LASSO calibration time, constrained optimization time, and total
23 end-to-end digital-twin update time, demonstrating compatibility with the one-hour control
24 interval. We also revised the conclusion to identify longer-duration and full-scale deployment
25 studies as future work.
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